Classifying *Pygoscelis* penguins' sex from four body measurements: a comparison of nearest-neighbour classifiers

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Abstract

Since the dawn of time, humans have wondered whether it is possible to tell whether a penguin is male or female based only on body size and length of bill. We consider this urgent question. We find that normalising the measurements matters a great deal, that the choice of distance metric does not matter at all, and that adding the categorical columns such as island of origin makes matters slightly worse.

1 Introduction

Since the dawn of time, humans have wondered whether it is possible to tell whether a penguin is male or female based only on body size and length of bill. We consider this urgent question. We find that normalising the measurements matters a great deal, that the choice of distance metric does not matter at all, and that adding the categorical columns such as island of origin makes matters slightly worse.

2 Data and method

According to Horst et al., body size measurements were collected for "three species of *Pygoscelis* penguins that breed on islands throughout the Palmer Archipelago, Antarctica" [Horst et al., 2022]. The original data was published as Horst et al. [2020]. Three hundred and forty-four penguins were measured, of which 333 were also sexed, i.e. their sex was determined.

In our methodology, the remaining eleven are dropped, because a missing label cannot be imputed. We hold out 20% of the sexed birds as a test set, touch it once, and do everything else — choosing k, the metric, whether to normalise and which columns to use — on the remaining 266 by five-fold cross-validation repeated six times, giving 30 train/test splits per configuration.

3 Results

Table 1 gives the models we have to beat. A classifier that always answers with the commoner label scores 0.504, which is the number every other number in this paper should be read against.

Table 2 gives the full factorial, and Figure 1 the same numbers with their error bars.

4 Conclusions

It is now time to end this paper.

setting model	cats off, norm off	cats off, norm on	cats on, norm off	cats on, norm on
Dummy (most frequent)	0.504	0.504	0.504	0.504
Decision tree	0.858	0.858	0.865	0.866
Random forest	0.882	0.880	0.897	0.895

Table 1: Accuracy of three reference models, averaged over 25 train/test splits. The two tree models are unaffected by normalisation, as they must be: a tree compares one feature against a threshold, and rescaling a column cannot change which side of it a bird falls.

metric	setting k	cats off, norm off	cats off, norm on	cats on, norm off	cats on, norm on
chebyshev	1	0.734	0.860	0.734	0.872
	3	0.764	0.881	0.764	0.885
	5	0.771	0.889	0.771	0.897
	9	0.785	0.904	0.785	0.902
	15	0.789	0.911	0.789	0.890
	25	0.784	0.913	0.784	0.819
	45	0.783	0.862	0.783	0.695
euclidean	1	0.747	0.860	0.746	0.865
	3	0.787	0.893	0.787	0.897
	5	0.780	0.900	0.781	0.897
	9	0.788	0.910	0.789	0.902
	15	0.787	0.905	0.788	0.894
	25	0.787	0.902	0.787	0.834
	45	0.788	0.878	0.788	0.817
manhattan	1	0.757	0.864	0.764	0.858
	3	0.803	0.889	0.802	0.892
	5	0.791	0.900	0.789	0.896
	9	0.789	0.910	0.790	0.898
	15	0.791	0.908	0.791	0.889
	25	0.788	0.901	0.788	0.839
	45	0.788	0.891	0.789	0.822

Table 2: Accuracy of our nearest-neighbour classifier at every combination of the four factors, averaged over 30 train/test splits. Normalisation is worth roughly thirteen points; the three metrics are quite similar overall; accuracy peaks around $k = 9 - 15$ and falls away on either side.

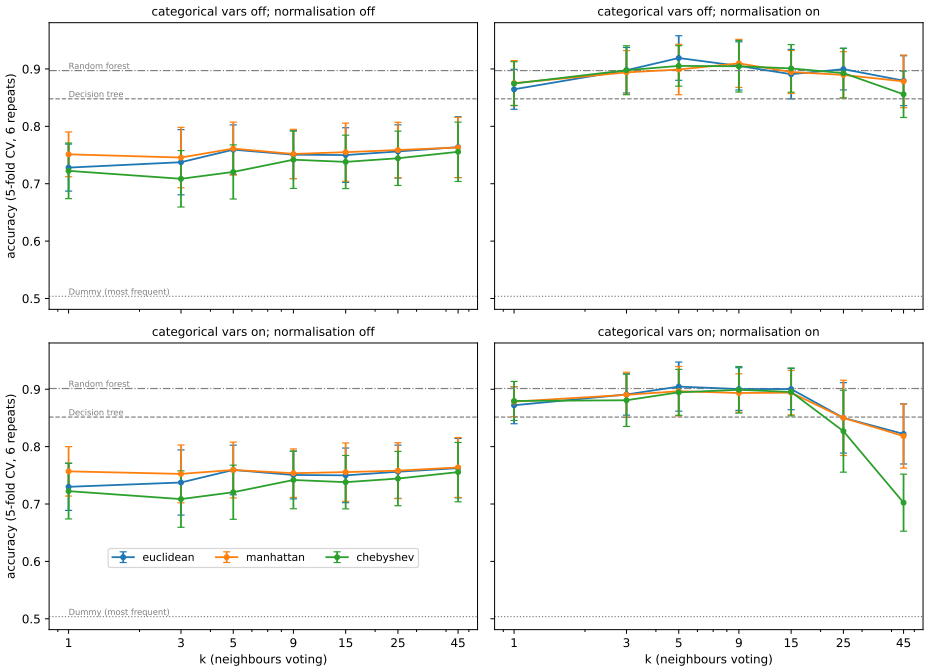


Figure 1: Accuracy against the number of voting neighbours k, for each combination of feature set and normalisation, with one line per distance metric. Error bars are one standard deviation across the 30 splits; horizontal lines mark the reference models of Table 1. Note that the categorical columns change nothing while the features are unnormalised — a zero-or-one column contributes almost nothing to a distance dominated by body mass in grams — and start doing harm as soon as they are given equal weight.

5 Ethics statement

No penguins were harmed during this computational experiment, but we are not entirely sure about the original data-gathering.

References

Allison M Horst, Alison Presmanes Hill, and Kristen B Gorman. Palmer Archipelago Penguins data in the palmerpenguins R package - an alternative to Anderson's Irises. *R Journal*, 14(1): 244, 2022.

Allison Marie Horst, Alison Presmanes Hill, and Kristen B Gorman. *palmerpenguins: Palmer Archipelago (Antarctica) penguin data*, 2020. URL `https://allisonhorst.github.io/palmerpenguins/`. R package version 0.1.0.